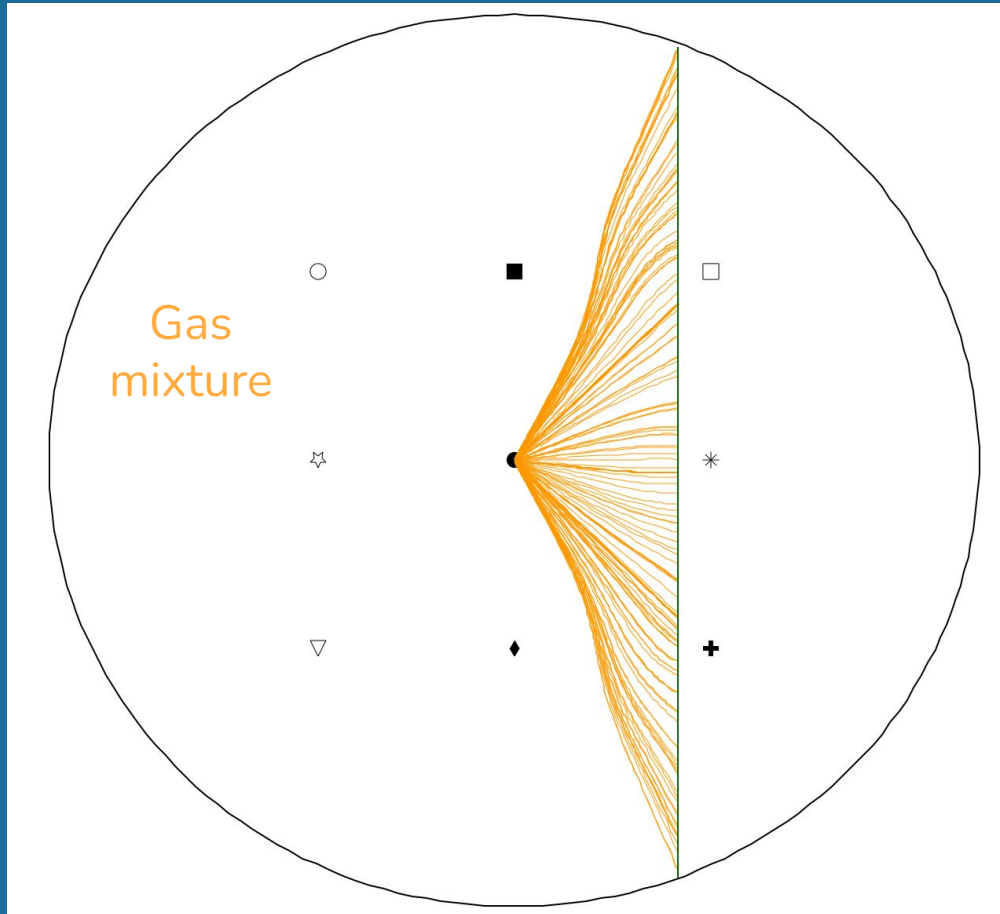

Garfield++ Drift Chamber Simulations

Jay Robertson



Drift cell - a sense wire surrounded by a square of eight grounded wires

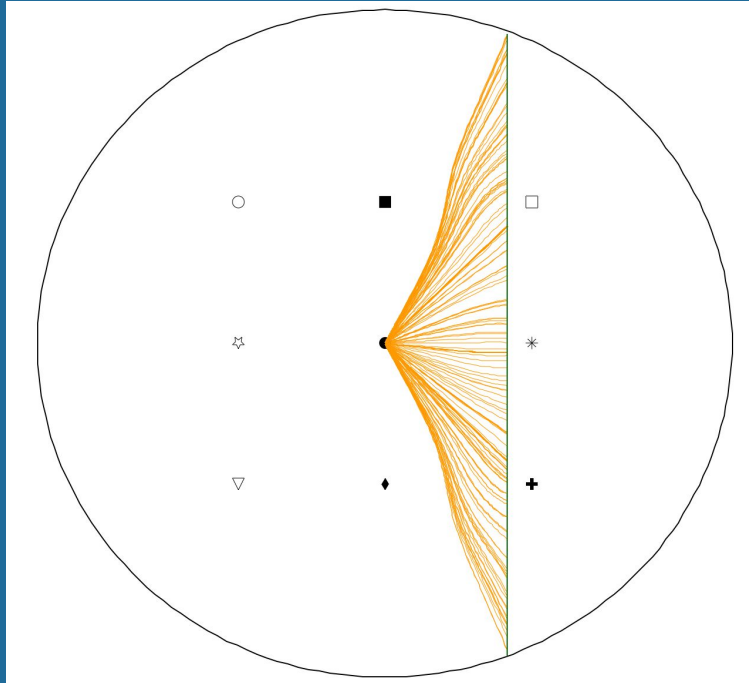
Track - path of a particle (muon used as example) through the drift cell

Drift line - trajectory of ions from their ionization point on the track to the sense wire

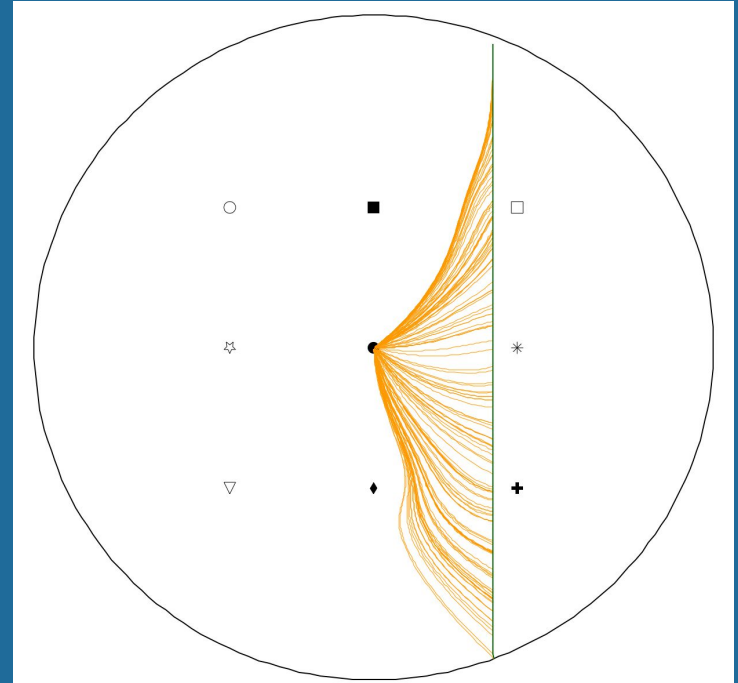
Cluster - a set of drift lines originating at the same location on the track

Drift tube – identical single track

B field = 1.5T



Without B field



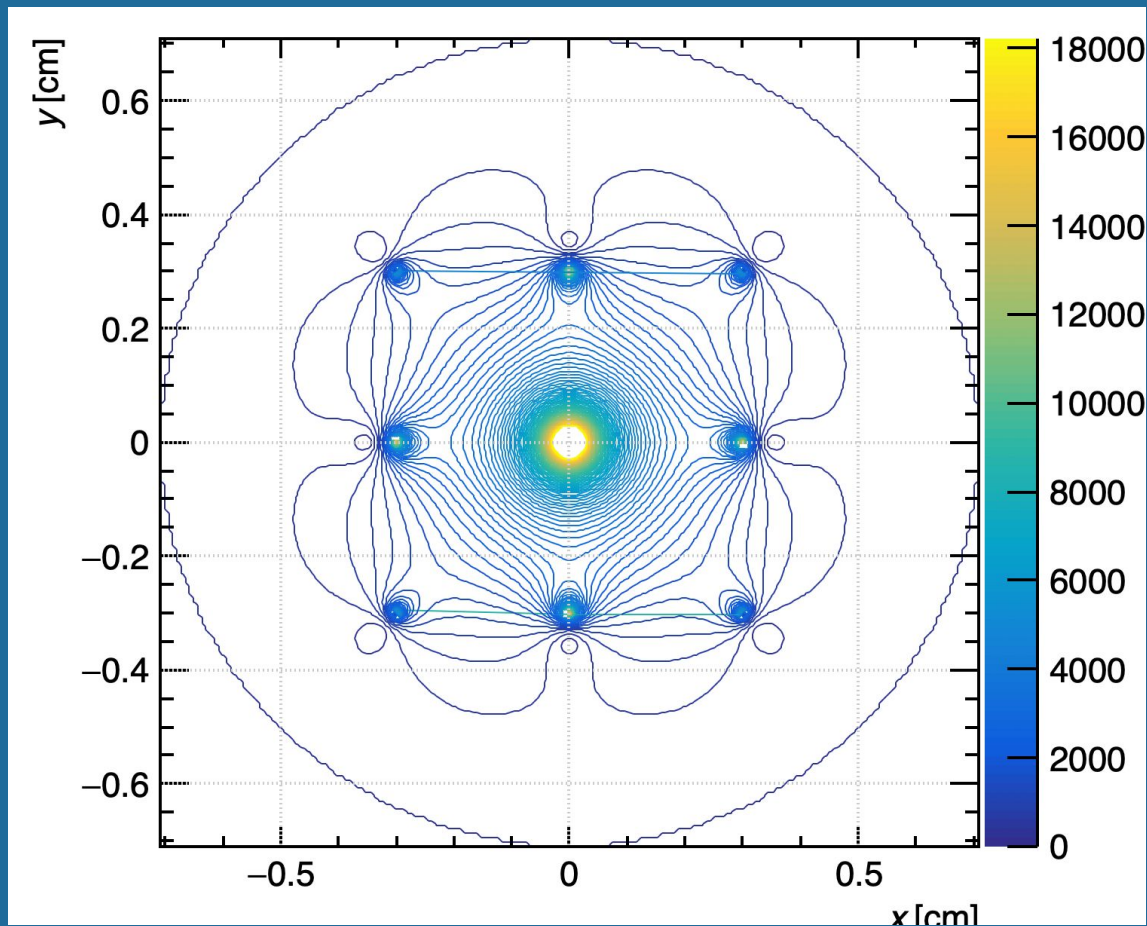
With B field

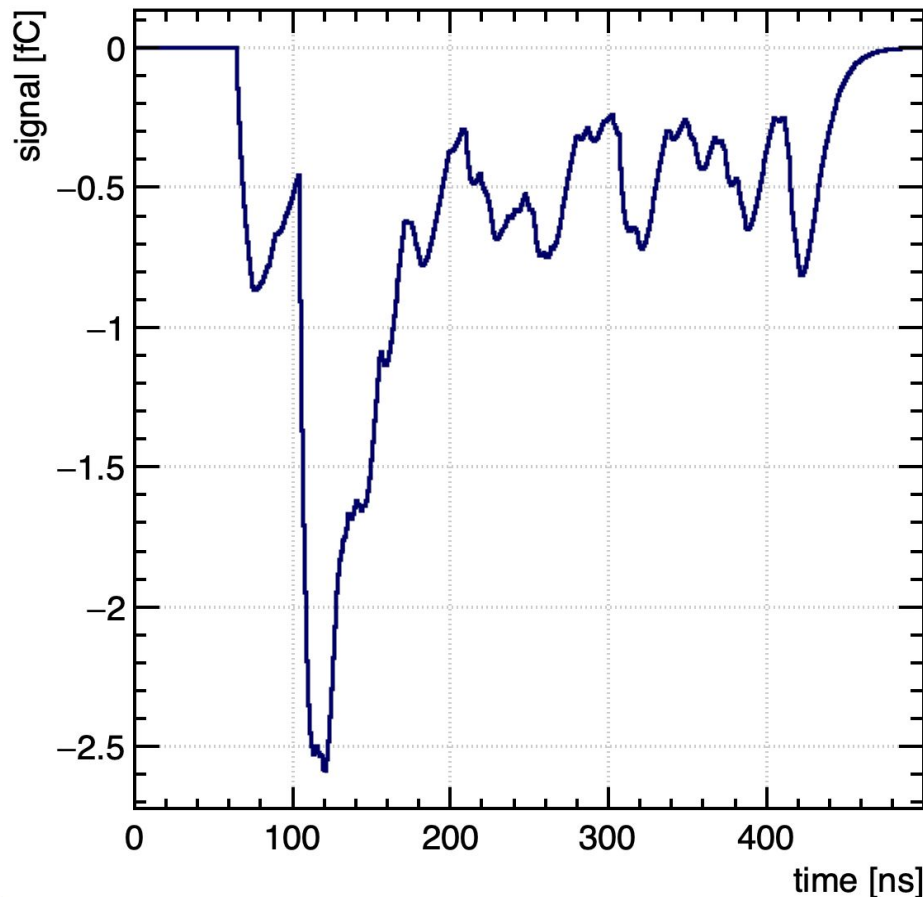
Electric field

Ground wires do contribute to electric field

Stronger electric field closer to the sense wire

Field is lower outside of the drift cell because there are no neighbouring cells





Signal

Response function given by:

$$f(t) = (t/\tau) * e^{-t/\tau}$$

$$\tau=8$$

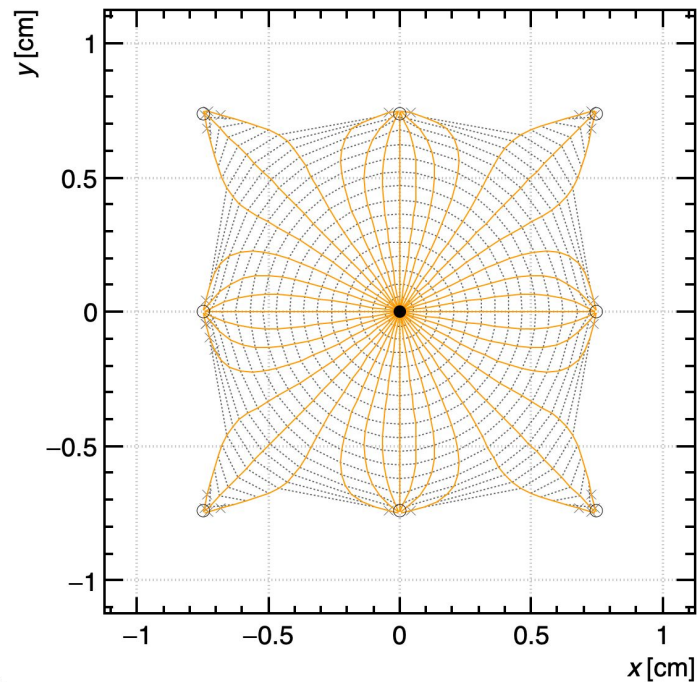
The signal from each ionization event

Each peak of the signal corresponds to a cluster

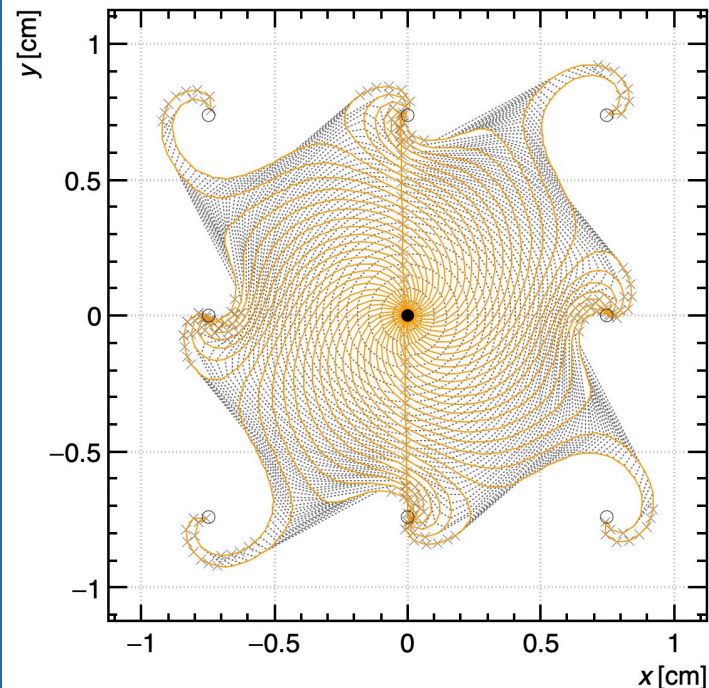
Isochrons – circular sampling

Isochron - lines of constant drift time, lines electrons reach at the same time

Without B field



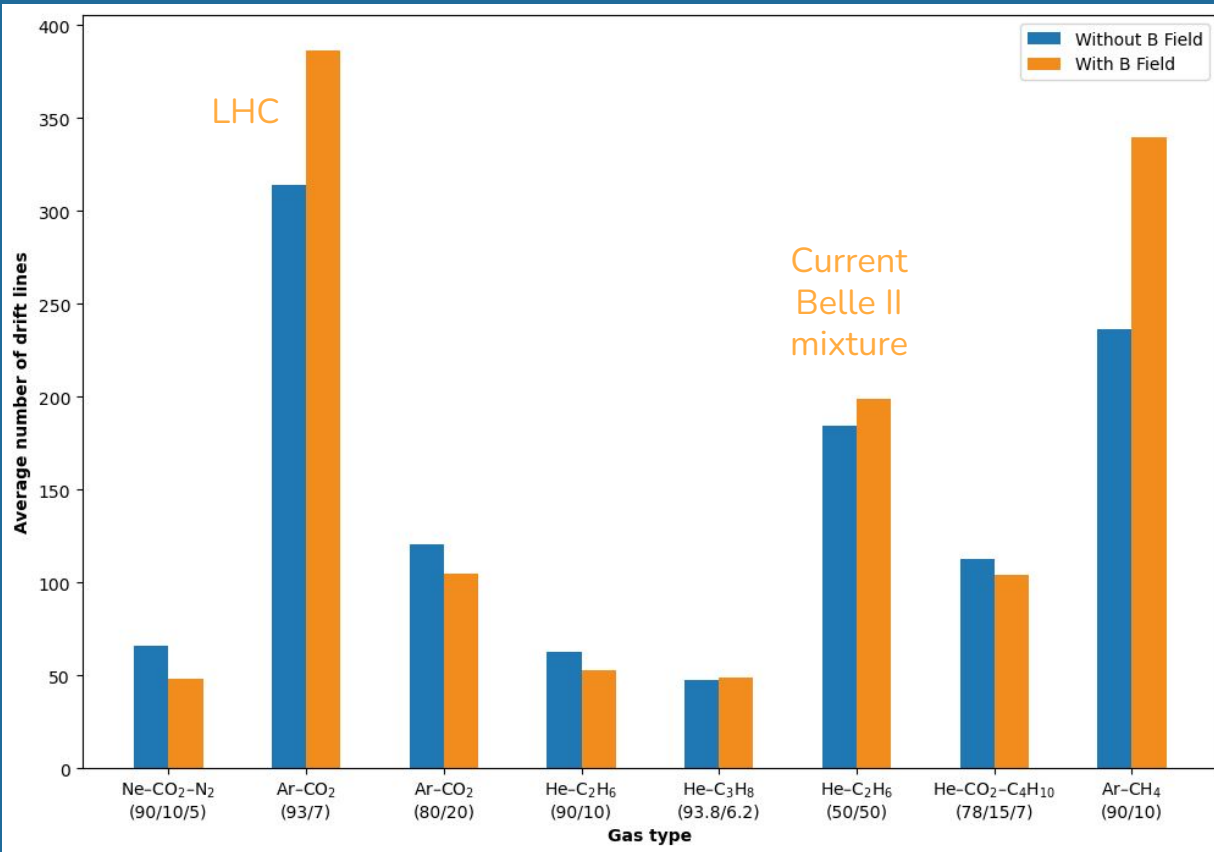
With B field



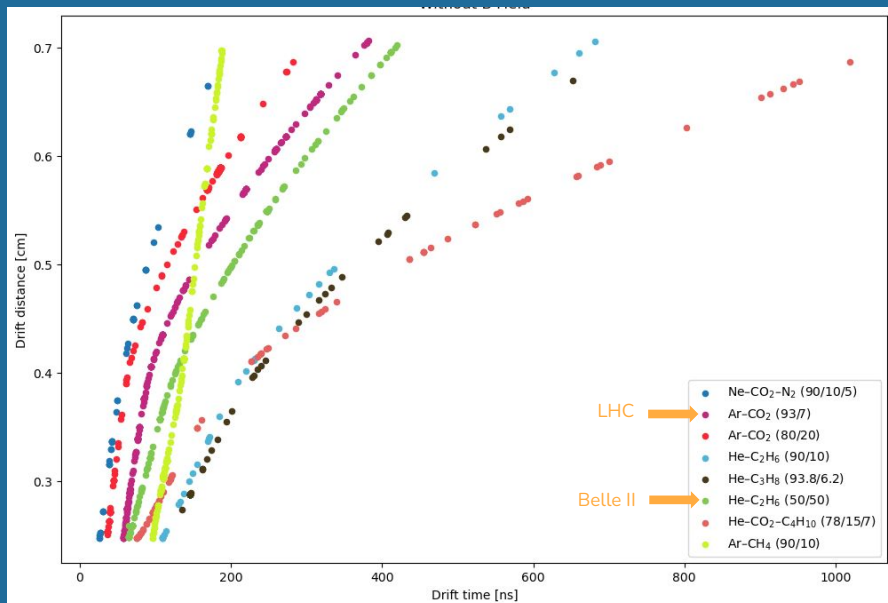
Gas mixtures

Argon gas mixtures generally resulted in more ionization events

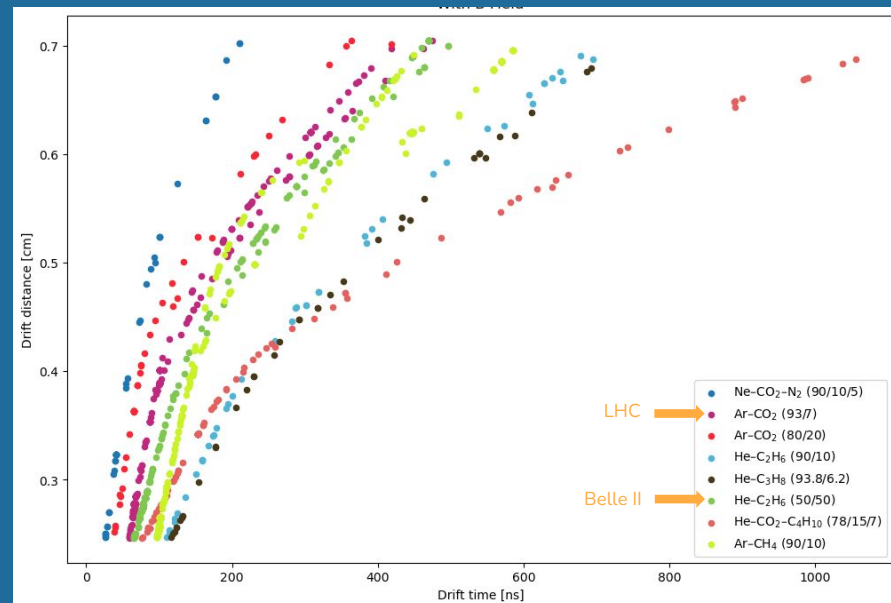
This is for a fixed trajectory as the number of drift lines will change based on track trajectory



Drift velocity in different gas mixes



Without B field



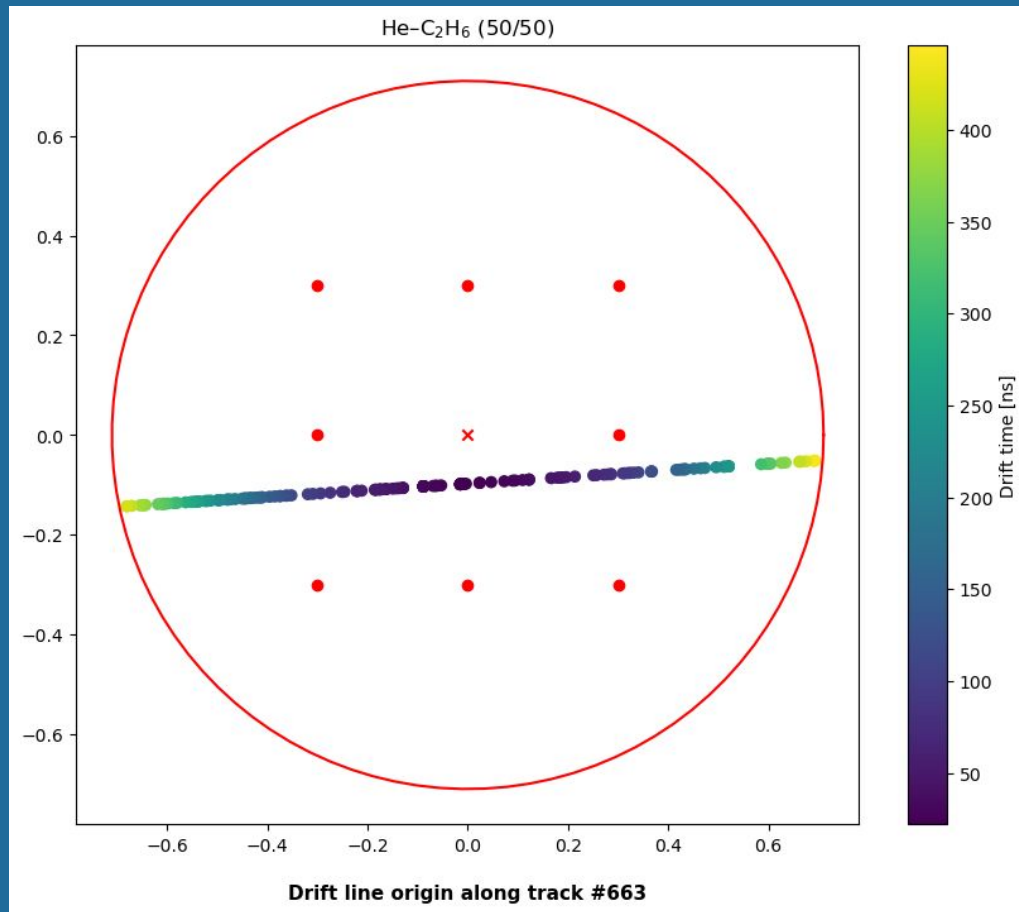
With B field

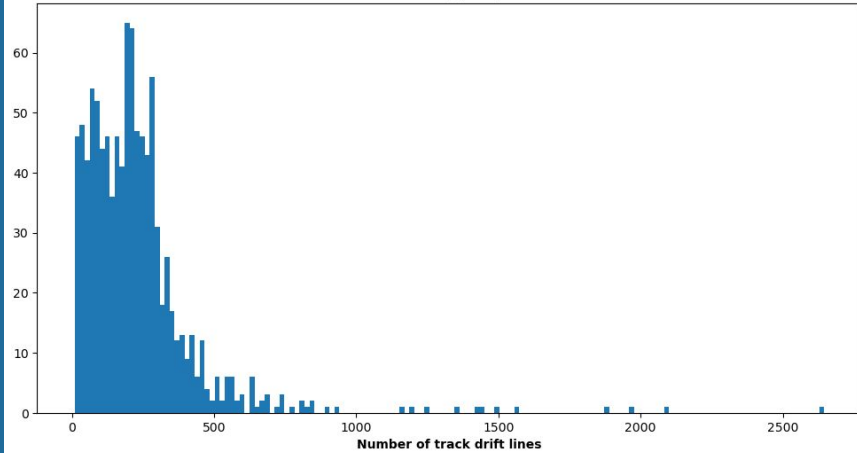
Random tracks

Randomly generated tracks
at different starting
positions and angles

Alg chose two points along
tub circumference and
plotted a vector track
between them

Ran script for two different
gas mixtures (1000 tracks
each)



Ar-CO₂ (93/7)

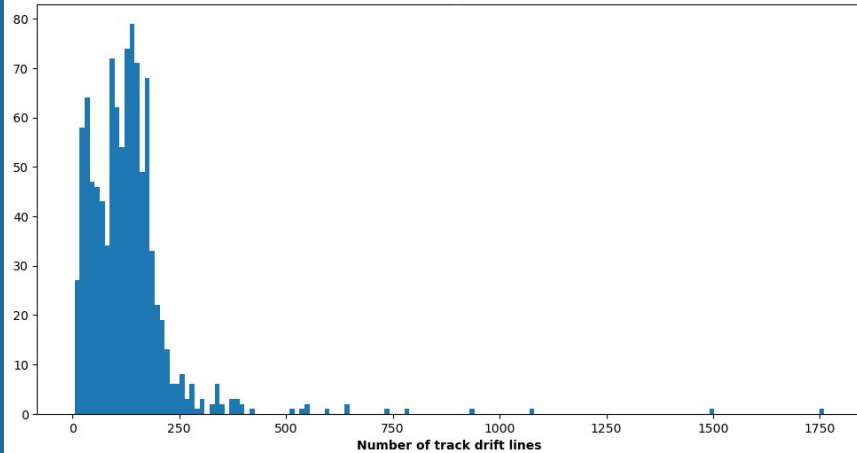
Number of drift lines

See distinct differences in the number of drift lines in each gas

Confirms that gas choice has significant effect on ionization rates in this simulation as expected

No cases of 0 drift lines, but see results of as few as 5

Few cases of exceedingly large numbers of drift lines

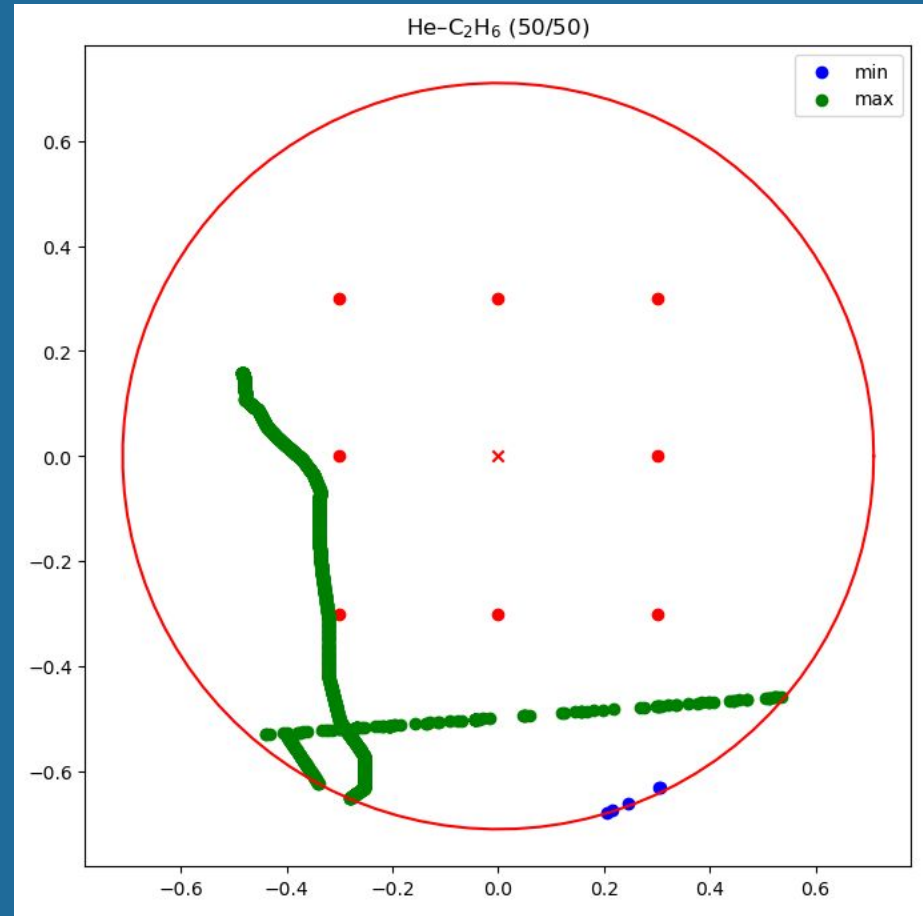
He-C₂H₆ (50/50)

Drift line number outliers

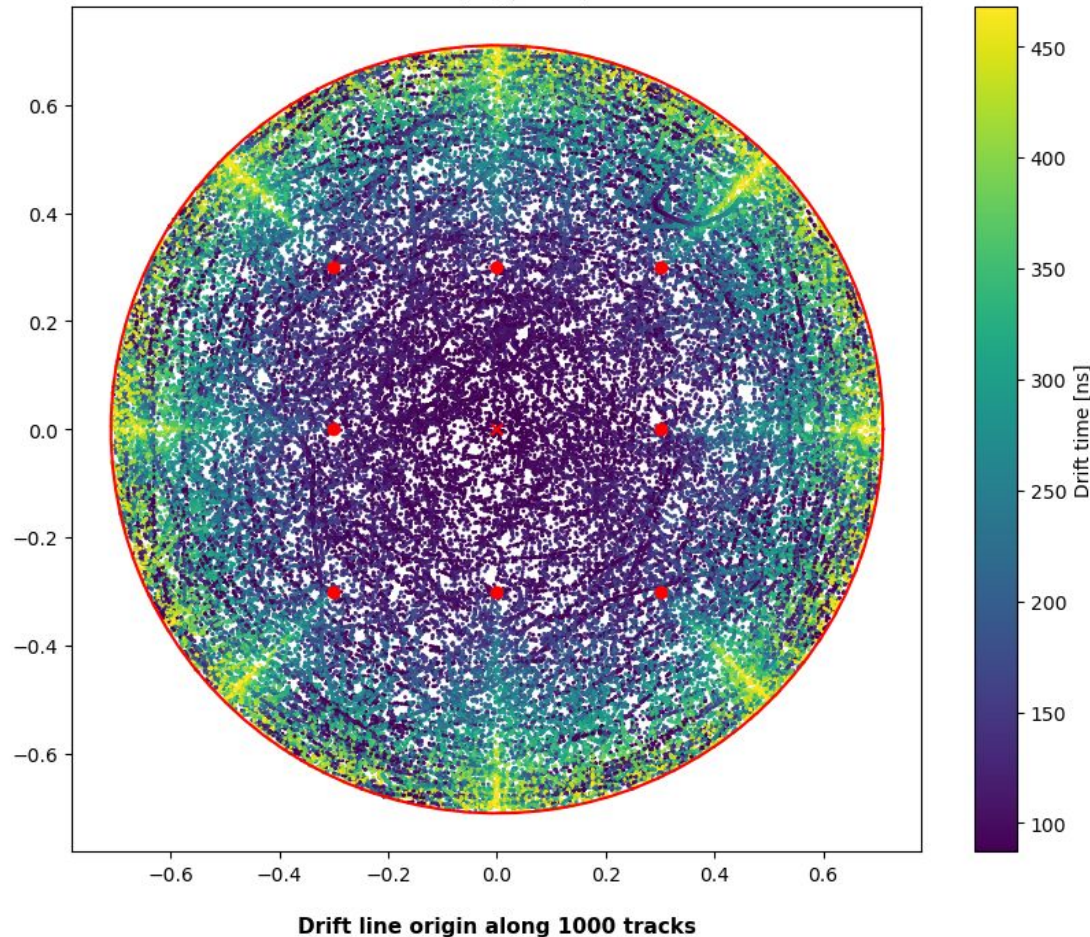
These are the two tracks with the shortest and longest lengths in the 1000 randomly generated

Smaller track length = less instances of muon interacting with gas mixture

Rarely, a particle is ejected from the linear track and moves around the cell, inducing excess ionization events



He-C₂H₆ (50/50)



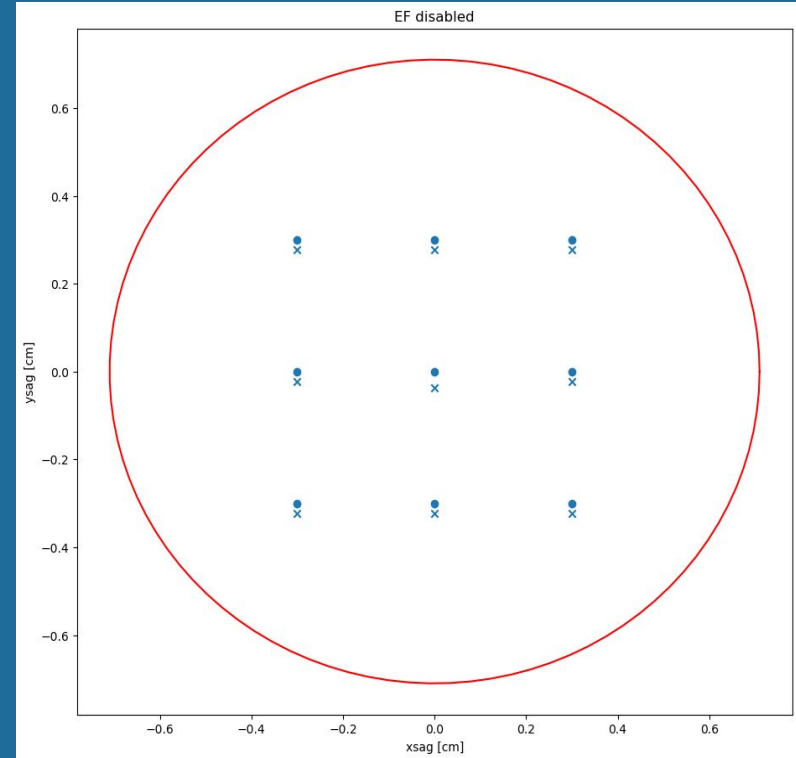
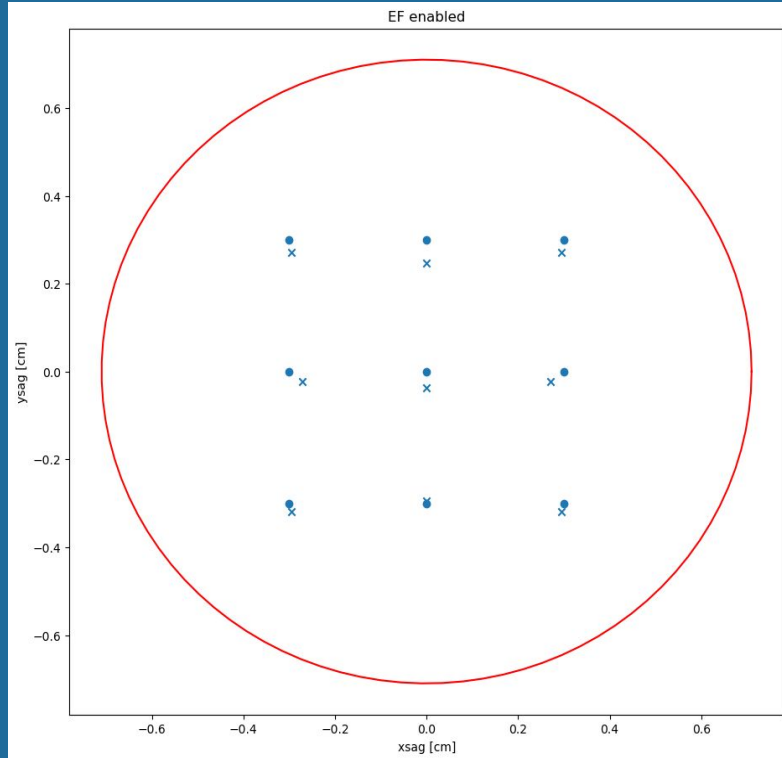
All tracks

Further away from sense wire = longer to travel to get there

Regions of shadowing and large drift times come from fact that we only have one cell

Sag due to forces acting on wires

● = original wire location
x = sagged wire location



g



E_f = electrostatic force